

IEEE 14-Bus GFL↔GFM Switching with Project AGSI++

Production all-KCL 200-s switching and SG-reference handback

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Abstract

One six-state EMF6 synchronous generator (SG) and four registered full-state dual-mode IBRs are simulated by the project-owned fixed-step, all-KCL hybrid engine for 200 s. The exact chronology is SG trip at 20 s, load increase at 50 s, bus-9 fault at 85–85.15 s, line 6–13 trip at 110 s, topology restoration and SG-close request at 145 s, and endpoint at 200 s. The operating point uses a 1.00 pu slack reference with a $\pm 5\%$ admissible voltage band. The run converges with maximum accepted-step residual 9.99×10^{-9} and at most one subdivision level. The synchronism guard accepts the SG at 146.825 s; all IBRs then return from GFM to GFL through the separately declared coordinated SG-reference handback. By the 200 s endpoint the SG frequency has settled to 60.000 Hz, though five remote load buses remain slightly below the $\pm 5\%$ voltage band. No Bayesian-optimisation controller is used. A seeded band-limited display ripple is drawn beside, not instead of, each raw trace and affects figures only.

1 Case, decision index, and switching semantics

The system base is 100 MVA and 60 Hz. The SG is at bus 1; IBR1–IBR4 are at buses 2, 3, 6, and 8. The buses carry a parallel bus_role label (REF/PV/PQ/GFM/GFL) that is *presentation-only* and never feeds the power-flow equations (Section 1): the SG bus 1 is REF, buses 2/3/6/8 (the IBRs) are GFL resources on this operating point, and the remaining load buses are PQ. The numerical internal bus type (REF= 1, PV= 2, PQ= 3) in the 12-column bus_data is unchanged by this descriptor. The index that produced the results in this report is the seven-term AGSI++ average

$$J_V = \frac{|V - V_{ref}|}{0.10}, \quad J_f = \frac{|f - 60|}{0.50}, \quad J_R = \frac{|df/dt|}{1.0}, \quad (1)$$

$$J_P = \frac{|P_{ref} - P|}{0.20}, \quad J_{SCR} = \max(0, 3/SCR - 1), \quad J_{lock} = \frac{|v_q|}{0.10}, \quad (2)$$

$$J_{GRA} = 1 - GRA, \quad AGSI_i^{++} = \min\left(1, \max\left(0, \frac{1}{7} \sum J_k\right)\right), \quad (3)$$

where $GRA = 1$ when an online SG or committed GFM supplies a reference. Its first four bases are source-guided; the equal $1/7$ weighting of the remaining three terms is PROJECT_DERIVED and is the part of (3) whose provenance cannot be defended term by term. Section 1.1 states the redesigned index that replaces it; the numbers in this report predate that redesign and were

produced by (3). A design-time weight tuner exists in the repository at `scripts/reporting/optimize_agsi_weights.m` (NUMERICAL_METHOD, random search plus coordinate descent). Its output was *not* used: the reported run uses equal 1/7 weights.

The generic local state machine uses $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$, $T_{d,on} = 0.10$ s, and $T_{d,off} = 1.00$ s. In this declared chronology, however, the SG-trip transaction is a CASE_DEFINED all-GFM override, and the return at SG close is a coordinated handback. Neither transaction is presented as an index-threshold crossing.

1.1 Redesigned decision index: severity plus hard gates

The switching decision is about restoring a voltage and frequency reference, so the redesign keeps only those two quantities in the continuous objective and demotes everything else to an admissibility test. The severity is

$$S_i = \min(1, \max(0, w_V J_V + w_f J_f)), \quad J_V = \frac{|V - V_{ref}|}{\Delta V_{base}}, \quad J_f = \frac{|f - f_0|}{\Delta f_{base}}, \quad (4)$$

with $\Delta V_{base} = 0.10$ pu, $\Delta f_{base} = 0.50$ Hz, and $w_V = w_f = 0.5$, $w_V + w_f = 1$. The two bases are SOURCE_DEFINED; the equal split follows the source ratio 0.30 : 0.30 renormalised, so no free weight remains.

Every other former term becomes a hard not-to-exceed gate returning $L \in \{0, 1\}$, never a weighted cost:

Table 1: Admissibility gates. A mode change requires every gate admissible.

Gate	Test	Meaning	Classification
RoCoF	$ df/dt \leq 1.0 \text{ Hz s}^{-1}$	rate admissibility	SOURCE_DEFINED
Power tracking	$ P_{ref} - P \leq 0.20 \text{ pu}$	dispatch feasibility	SOURCE_DEFINED
Grid strength	$SCR \geq SCR_{crit}$	GFL model validity	SOURCE_DEFINED
PLL lock	$ v_q \leq 0.10 \text{ pu}$	synchronisation health	PROJECT_DERIVED
Reference	$GRA = 1$	a forming reference exists	PROJECT_DERIVED
Current	$ I \leq I_{max}$	converter limit	SOURCE_DEFINED

The switching law is then

$$\begin{aligned} \text{GFL} \rightarrow \text{GFM} : S_i &\geq \Gamma_{on} \wedge \bigwedge_k L_k, \text{ for } T_{d,on}, \\ \text{GFM} \rightarrow \text{GFL} : S_i &\leq \Gamma_{off} \wedge \bigwedge_k L_k, \text{ for } T_{d,off}, \end{aligned} \quad (5)$$

retaining the existing hysteresis and dwell values. Each weight is now either source-defined or fixed by normalisation, and each remaining quantity is a published limit rather than a tuned coefficient. This design is frozen here *before* any re-run; equations (4)–(5) are not used to produce any number in Sections 4–6 of this report.

1.1.1 Why J_V, J_f suffice as severity, and why the weights are 0.5:0.5

So that the design can be defended when asked “why only J_V, J_f ” and “why these weights,” the reasoning is stated in three parts.

(1) Why J_V and J_f suffice as the severity. The objective of switching is the single question “*is the reference healthy?*”, and the two quantities a grid-forming reference directly regulates

are the voltage magnitude $|V|$ and the frequency f (voltage through the voltage PI and the Q - V droop; frequency through the virtual rotor and the droop). The severity therefore has to measure “distance from the reference” with exactly these two quantities. Every other term of (3) is not a reference-severity but a condition of the surrounding system.

(2) Why the other five terms must be gates, not weights. A condition such as “ SCR must be sufficient,” “current must stay below I_{max} ,” or “the PLL must be locked” is a pass/fail statement, not a “distance” that can be averaged with other terms: there is no physics in which a good SCR compensates for an over-current. Making them weighted terms lets a good value *mask* a bad one, and worse, the averaging makes the index insensitive. A concrete count from (3): with the $1/7$ average a single term can trigger the threshold $\Gamma_{on} = 0.65$ only once it reaches $7 \times 0.65 = 4.55$, i.e. $J_V = 4.55$ back-transforms to $|\Delta V| = 4.55 \times 0.10 = 0.455$ pu — a catastrophic voltage collapse — yet the $1/7$ index reads it merely as “at threshold.” The index therefore cannot distinguish “one severe term” from “all terms mildly stressed,” and does not say what is actually wrong. The two-term severity always saturates at 1, so the threshold 0.65 reads directly as “the normalized deviation reaches 65% of the band,” which is interpretable at a glance.

(3) Why the weights are 0.5:0.5. Separate the bases from the weights. The bases ($\Delta V_{base} = 0.10$ pu and $\Delta f_{base} = 0.50$ Hz) make J dimensionless and comparable: $J_V = 1$ means a full 10% band deviation and $J_f = 1$ means a full 0.5 Hz band deviation. The weight then states whether the two normalised deviations are equally important. The source EECN49-P4 Sec. 4.1 assigns the weight ratio of V to f as 0.30:0.30, i.e. it treats them as equally important; renormalising to unit sum gives exactly 0.5:0.5. **The weights are therefore pinned by normalisation from the source, not tuned.** If asked “why not 0.7:0.3,” the answer is: changing them would need source support, and the source treats V and f as equal. This is the provenance difference from the legacy $1/7$ weights: $1/7$ has no source support and cannot be defended, whereas 0.5:0.5 can be defended to the end.

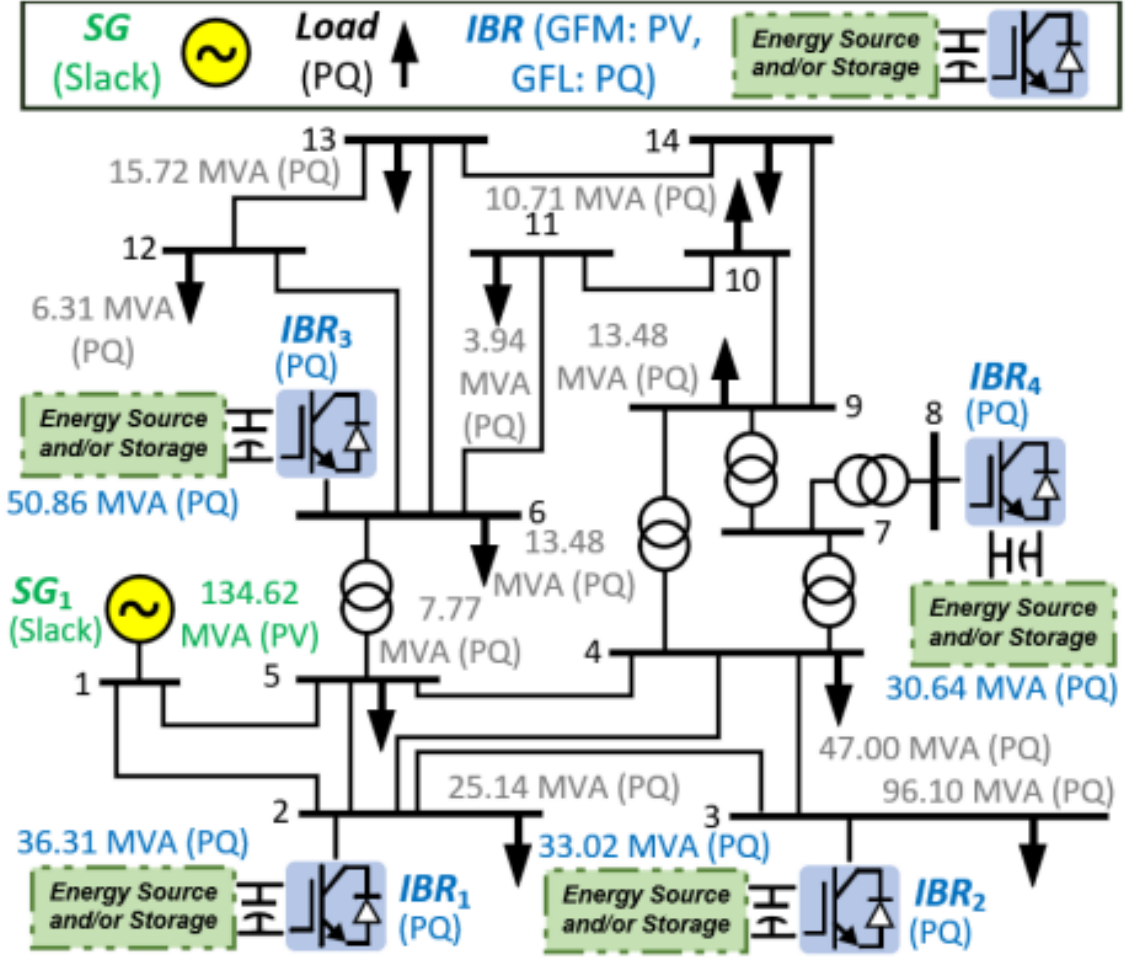


Figure 1: Network and resource locations used by the simulation.

2 Dynamic equations

The operational SG state is $x_{SG} = [\delta, \Delta\omega, E'_q, E'_d, E''_q, E''_d]^T$:

$$\dot{\delta} = \omega_0 \Delta\omega, \quad 2H\Delta\dot{\omega} = P_m - P_e - D\Delta\omega, \quad (6)$$

$$T'_{d0}\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \quad T'_{q0}\dot{E}'_d = c_q E''_d - d_q E'_d, \quad (7)$$

$$T''_{d0}\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)I_d, \quad T''_{q0}\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)I_q. \quad (8)$$

The source singular limit $T'_{q0} = 0$ freezes $E'_d = 0$ exactly. Two prime-mover states remain outside the six-state electromagnetic order:

$$T_{SV}\dot{P}_{SV} = -P_{SV} + P_c - \Delta\omega/R, \quad T_{CH}\dot{P}_m = -P_m + P_{SV}.$$

They provide offline runback/phase preparation and post-close primary regulation. The numerical mapping is diagnostic and is not a machine-specific governor validation.

2.1 IBR state superset

Each IBR carries a constant-dimension **20-state** switching superset: the 13-state grid-forming branch (REGFM_B1, generator model G2) and the 7-state grid-following branch (the WECC

REGC_A and REEC_A pair) exist at all times,

$$x_{GFM} = [\Delta\omega_m, \delta_{IT}, x_w, x_E, \delta_{PLL}, x_{PLL}, P_{inv,f}, I_{d,f}, Q_{inv,f}, V_{inv,f}, I_{q,f}, \delta_{IT,max}, \delta_{IT,min}]^T, \quad (9)$$

$$x_{GFL} = [V_{t,f}, P_f, I_{q,cmd,f}, P_{ord}, V_{LVPL,f}, I_{p,reg}, I_{q,reg}]^T, \quad (10)$$

so that $x_{IBR} = [x_{GFM}^T \ x_{GFL}^T]^T \in \mathbb{R}^{20}$. All quantities are per-unit on the device base M_{base} ; the network interface applies $\kappa = S_{base}/M_{base}$. Powers are positive into the network. Only the branch selected by the present mode q contributes a nonzero derivative; the other branch satisfies $\dot{x} = 0$ exactly.

2.2 Grid-forming branch: 13 states, term by term

The angle reference is the sum of a phase-locked component and a virtual-machine component, so the GFM branch *does* own a PLL pair. With $V = V_x + jV_y$ the terminal voltage and $v_q = -V_x \sin \delta_{PLL} + V_y \cos \delta_{PLL}$ its quadrature error,

$$\dot{x}_{PLL} = v_q, \quad \dot{\delta}_{PLL} = \omega_0 (k_{p,PLL} v_q + k_{i,PLL} x_{PLL}), \quad (11)$$

both frozen to zero while $|V| < V_{PLL,frz}$ so that a collapsed terminal cannot drive the tracker. The virtual synchronous machine supplies the frequency deviation, with a washout state providing transient-only damping:

$$2H\Delta\dot{\omega}_m = P_{ref,inv} - P_{inv,f} - (1/m_p + D_1) \Delta\omega_m - D_2 (\Delta\omega_m - x_w), \quad (12)$$

$$\dot{x}_w = \omega_D (\Delta\omega_m - x_w). \quad (13)$$

Here $1/m_p$ is the droop gain, D_1 the steady damping, and D_2 acts only on the high-pass component $\Delta\omega_m - x_w$. The internal angle is stored *relative* to the PLL, which is what keeps the two angle sources consistent:

$$\dot{\delta}_{IT} = \omega_0 \Delta\omega_m - \dot{\delta}_{PLL}, \quad \delta_{VSM} = \delta_{PLL} + \delta_{IT}, \quad (14)$$

with $\dot{\delta}_{IT}$ held at zero whenever δ_{IT} sits on the active bound of $[\delta_{IT,min}, \delta_{IT,max}]$ and the derivative would push it further out. The voltage magnitude comes from a PI regulator with reactive droop and anti-windup,

$$\dot{x}_E = V_{ref} - V_{inv,f}, \quad E_{VSM}^{raw} = V_{ref} - m_q Q_{inv,f} + k_{pv} (V_{ref} - V_{inv,f}) + k_{iv} x_E, \quad (15)$$

where $E_{VSM} = \text{sat}(E_{VSM}^{raw}, E_{min}, E_{max})$ and $\dot{x}_E$ is set to zero when the limiter is saturated in the direction of further integration. The limits E_{min} and E_{max} are *not constants*: they are recomputed every step from the current state so that the q -axis current cannot exceed its limit,

$$E_{min} = \sqrt{(V_{inv,f} - I_{q,max}^{SS} X_L)^2 + (I_{d,f} X_L)^2}, \quad (16)$$

$$E_{max} = \sqrt{(V_{inv,f} + I_{q,max}^{SS} X_L)^2 + (I_{d,f} X_L)^2}, \quad (17)$$

with the current limit split by PQFlag: on d -axis priority $I_{d,max}^{SS} = k_f I_{max}^{SS}$ and

$$I_{q,max}^{SS} = \sqrt{\max((I_{max}^{SS})^2 - I_{d,f}^2, 0)},$$

and the reverse assignment on q -axis priority. The x_E anti-windup uses the same conditional-hold rule: $\dot{x}_E = 0$ when E_{VSM}^{raw} is outside the band and the derivative still pushes outward. The current-angle bounds are themselves states, integrating the mismatch between the steady-state d -axis current limit and the measured d -axis current, so that the angle authority contracts when the device is overloaded:

$$\dot{\delta}_{IT,max} = k_I \left(I_{d,max}^{SS} - I_{d,f} \right) \in [0, \delta_{max}], \quad \dot{\delta}_{IT,min} = k_I \left(-K_e I_{d,max}^{SS} - I_{d,f} \right) \in [-\delta_{max}, 0], \quad (18)$$

each frozen on its own bound by the same conditional-hold rule, with $\delta_{max} = \arcsin(X_L I_{max}^{SS})$ and $\dot{\delta}_{IT,min} = 0$ when the energy-storage flag is inactive. The five remaining states are first-order measurement filters,

$$T_{pf} \dot{P}_{inv,f} = \kappa P - P_{inv,f}, \quad T_{If} \dot{I}_{d,f} = \kappa I_d - I_{d,f}, \quad T_{Qf} \dot{Q}_{inv,f} = \kappa Q - Q_{inv,f}, \quad (19)$$

$$T_{Vf} \dot{V}_{inv,f} = |V| - V_{inv,f}, \quad T_{If} \dot{I}_{q,f} = \kappa I_q - I_{q,f}, \quad (20)$$

where $P + jQ = V \overline{I_{out}}$ and (I_d, I_q) are the injected current projected on the PLL frame. The network injection is voltage-behind-impedance with circular saturation:

$$I_{out} = \begin{cases} I_{unc}, & |I_{unc}| < I_{max}^F / \kappa, \\ \frac{I_{max}^F}{\kappa} \frac{I_{unc}}{|I_{unc}|}, & \text{otherwise,} \end{cases} \quad I_{unc} = \frac{E_{VSM} e^{j\delta_{VSM}} - V}{\kappa (R_e + jX_L)}. \quad (21)$$

2.3 Grid-following branch: 7 states, term by term

The GFL branch is the registered WECC pair and is a *current-source* model: it owns no PLL state because it aligns algebraically to the measured terminal-voltage angle. Its injection is

$$I = \frac{(I_{p,reg} \gamma(|V|) - jI_{q,reg}) e^{j\angle V}}{\kappa}, \quad (22)$$

with γ the REGC_A low-voltage-point gain and a high-voltage reactive clamp applied above V_{olim} . The states are three measurement filters, the power order, and the two regulated currents:

$$T_{rv} \dot{V}_{t,f} = |V| - V_{t,f}, \quad T_p \dot{P}_f = \kappa P - P_f, \quad (23)$$

$$T_{fltr} \dot{V}_{LVPL,f} = |V| - V_{LVPL,f}, \quad T_{pord} \dot{P}_{ord} = \text{sat}(\kappa P_{ref}, P_{min}, P_{max}) - P_{ord}, \quad (24)$$

$$T_{iq} \dot{I}_{q,cmd,f} = I_{q,cmd} - I_{q,cmd,f}, \quad (25)$$

where $\dot{P}_{ord}$ is additionally rate-limited to $[\dot{P}_{min}, \dot{P}_{max}]$. The REEC_A command pair divides the order by the filtered voltage and adds the voltage-dip reactive injection,

$$I_{p,0} = \frac{P_{ord}}{\max(V_{t,f}, 0.01)}, \quad I_{q,0} = \frac{\kappa Q_{ref}}{\max(V_{t,f}, 0.01)} + I_{q,inj}, \quad (26)$$

$$I_{q,inj} = \text{sat}(K_{qv} \text{db}(V_{ref,0} - V_{t,f}), I_{ql1}, I_{qh1}) \quad \text{active only if } V_{t,f} < V_{dip} \text{ or } V_{t,f} > V_{up}, \quad (27)$$

followed by the active-priority allocation

$$I_p = \text{sat}(I_{p,0}, 0, I_{max}), \quad |I_q| \leq \min\left(I_{max}, \sqrt{I_{max}^2 - I_p^2}\right).$$

The regulated currents then track their commands through the converter lag with the sourced ramp limits,

$$T_g \dot{I}_{p,reg} = I_p^{tgt} - I_{p,reg} \quad (\dot{I}_{p,reg} \leq r_{rpwr}), \quad (28)$$

$$T_g \dot{I}_{q,reg} = I_{q,cmd,f} - I_{q,reg} \quad (\dot{I}_{q,reg} \in [I_{qr,min}, I_{qr,max}]), \quad (29)$$

where I_p^{tgt} is I_p passed through the LVPL characteristic when that switch is enabled and floored at zero below V_{lvpl0} . The 0.01 pu division guard is NUMERICAL_METHOD; every gain, time constant, and limit above is SOURCE_DEFINED.

2.4 Mode dispatch

In GFL mode the active indices are 14:20 and $\dot{x}_{1:13} = 0$; in GFM mode the active indices are 1:13 and $\dot{x}_{14:20} = 0$; in the tripped mode all twenty derivatives are zero and the injection is identically zero. Only the active branch enters the dynamic solve, and the active-index set is queried from the device rather than hard-coded, so the layout above is not duplicated in the integrator. Atomic state mapping is followed by a full-KCL right-limit solve; a failed solve rolls the transaction back.

3 Switching formulation: what is solved at a mode change

This section states the mathematics executed at every switching instant. It is the governing formulation for the SG trip, the GFL↔GFM transfers, and the SG reclose.

3.1 The plant is a DAE, not an ODE

Between events the plant is a semi-explicit index-1 differential–algebraic system, not a plain ordinary differential equation:

$$\dot{x} = f_q(x, y, u), \quad (30)$$

$$0 = g_q(x, y, u) = Y_q V - I_{inj}(x, y). \quad (31)$$

Here x collects every device differential state solved inside the step (SG six electromagnetic states plus 4×20 IBR states, i.e. 86 states); the two prime-mover states of Section 2 are advanced by a separate analytic exponential update and enter (30) only through the input u , not as Newton unknowns. Accordingly u carries T_m and E_{fd} , y is the network state stored as rectangular bus voltages, and q is the *discrete mode*: the tuple of network topology, per-device mode, and breaker status. Equation (31) is the full all-KCL residual over all $2n_b$ real rows; a single module owns it, while each device contributes only its own slice of f_q and its own I_{inj} .

The discretisation is the implicit trapezoidal rule, so a normal step is not the explicit recursion $x_{k+1} = x_k + h f(\cdot)$ but the simultaneous nonlinear system

$$x_{k+1} - x_k - \frac{h}{2} [f_q(x_k, y_k, u) + f_q(x_{k+1}, y_{k+1}, u)] = 0, \quad (32)$$

$$g_q(x_{k+1}, y_{k+1}, u) = 0, \quad (33)$$

solved together for (x_{k+1}, y_{k+1}) . The EMF6 route uses a fixed corrector count of three. Both f_q and g_q carry the subscript q : they are the equations of the *present discrete mode*. A switching event changes q , hence changes the equations themselves.

3.2 The reference-role handover: SG to IBR

Before the mechanical detail below, four conceptual points must be settled, because this is where the formulation is most easily misunderstood: *the SG and the IBRs obey different equations, so how can an IBR “replace” the SG?*

3.2.1 The SG and the IBRs really are different DAEs

Confirmed at the outset: the state variables and the physical mechanisms differ completely, with no shared state at all:

$$\text{SG: } 2H \Delta\dot{\omega} = P_m - P_e - D\Delta\omega, \quad (34)$$

$$\text{GFM: } 2H_v \Delta\dot{\omega}_m = P_{ref,inv} - P_{inv,f} - \left(\frac{1}{m_p} + D_1 \right) \Delta\omega_m - D_2 (\Delta\omega_m - x_w), \quad (35)$$

The SG uses the real inertia H of the rotating mass, the real rotor angle δ , and the real mechanical torque P_m ; the GFM uses a *virtual* inertia H_v , the synthetic angle $\delta_{VSM} = \delta_{PLL} + \delta_{IT}$, and the setpoint $P_{ref,inv}$ — the two swing equations look similar but describe different physical quantities. The GFL branch has no swing equation at all; it is a pure current source. Hence *no equation sets* $\omega_{SG} = \omega_m$; each frequency variable belongs to its own device.

3.2.2 The reference is a role, not an equation transformation

In the time-domain DAE (Section 3, §6.1) **every bus is an unknown of the all-KCL residual** $0 = Y_q V - I_{bus}$ over all $2n_b$ real rows; no bus angle is pinned as an algebraic slack. The code never replaces a KCL row by a pinned angle, and `reference_owner_schema.m` states it directly: “reference ownership is supervisory/physical, NOT a KCL-row replacement.” The reference is therefore *not a bus type* but a role played by whichever device controls the island’s angle and frequency: while the SG is online the role belongs to the SG; while it is open it belongs to IBR1 (bus 2) through its GFM branch. What “transforms” at $t = 20$ s is **not the SG equations into IBR equations** but the handover of the reference-setting role from one device to another.

3.2.3 Three things change together at the handover instant

The mechanical detail lives in Section 3.5; at the role level, three changes happen inside a single transaction:

1. **SG:** the equations switch to the open-circuit mode — $I_d = I_q = 0$, $T_e = 0$, flux decay, the rotor coasts as $2H \Delta\dot{\omega} = T_m^{froz} - D\Delta\omega$, and *all six states stay in the state vector*
2. **IBR:** the reset map switches GFL→GFM by initialising the GFM branch so the *terminal current is continuous*, $I^+ = I^-$ — the inverse problem of Section 3.5
3. **Network:** the matrix Y changes (the SG breaker opens) and the KCL is re-solved for y^+

All three are committed together as one transaction, so there is no interval in which the island is “without a reference” (the supervisor plot keeps $GRA = 1$ throughout). The simultaneous GFM switch of all four IBRs at 20 s is therefore the result of this coordinated transaction, not of the four AGSI indices crossing their thresholds at the same instant.

3.2.4 H, ω, f differ — why are they interchangeable

This is the deepest point, and the answer is *we never equate them directly*. The coupling is not an equation between frequency variables; it is the network. Every device injects current into the same set of buses, so KCL forces the bus voltages into geometric consistency; at steady state all voltages rotate at the same frequency, and each device locks to that frequency by its own mechanism — the SG by its physical rotor (integrating $\dot{\delta} = \omega_0 \Delta\omega$), the IBR by a PLL driving $v_q \rightarrow 0$ to align with the terminal-voltage angle (the WECC GFL uses $\angle V$ directly, algebraically). *Steady-state synchronism is therefore emergent from the network, not a directly imposed condition.*

This is why the switch must be “continuous at the terminal”, not “continuous in the state”: the terminal current is what the network sees, so keeping I from jumping is exactly not disturbing the network’s balance. The internal variables (δ of the SG, ω_m, δ_{VSM} of the GFM) belong to each device and may be re-initialised freely as long as the terminal effect stays the same.

3.2.5 How the transient is handled

The post-handover transient is managed in three stages: (1) device level — the reset map finds x^+ such that the terminal current is continuous (Section 3.5); (2) system level — solve KCL by Newton with x^+ held fixed, for y^+ (Eq. (50)); (3) integrate forward from the consistent pair $(x(t_e^+), y(t_e^+))$. The transient then decays according to the device dynamics — droop, damping D_1, D_2 , the PLL, and the voltage PI — not according to any imposed condition.

The SG reclose differs because x^+ *cannot* be found this way: the SG angle δ is tied to a real rotating mass and cannot change instantaneously, so there is no freedom to invert an equation for a desired δ^+ . The roles are therefore reversed — instead of solving for the continuity condition, the system *waits* for the physics to bring it close to continuity itself (the synchronism gate, Eq. (54)). This is what makes the 145.000 s close request complete at 146.825 s.

3.3 The transition at a switching instant

Equations (32)–(33) are *not* continued across an event. At a switching instant t_e the trajectory undergoes a hybrid-DAE transition in three ordered stages:

$$q^- \longrightarrow q^+ \quad \text{discrete jump,} \quad (36)$$

$$x(t_e^+) = R_{q^- \rightarrow q^+}(x(t_e^-), y(t_e^-)) \quad \text{state reset map,} \quad (37)$$

$$0 = g_{q^+}(x(t_e^+), y(t_e^+), u) \quad \text{solve for } y(t_e^+) \quad \text{right-limit algebraic solve.} \quad (38)$$

Stage (36) selects the post-event admittance matrix Y_{q^+} directly from the declared topology set by its identifier, and commits the device-mode and breaker maps as a single transaction: the whole update is validated before any field is written, and a failed validation leaves the state identical to its input. Stage (37) is described below. Stage (38) solves *only* the algebraic block, by damped Newton iteration with a backtracking line search, holding x fixed at $x(t_e^+)$ and accepting the result only if the KCL residual meets the step tolerance; otherwise the whole transaction rolls back. Differential integration then resumes from $(x(t_e^+), y(t_e^+))$ at the next step.

Consequently two samples exist at t_e : a left limit and a right limit. The tables and figures in this report store the right limit.

3.4 The reset map is terminal continuity, not state copying

The key structural property is that **the state dimension never changes**. Each IBR is a constant-dimension 20-state superset in which the GFM states 1:13 and the GFL states 14:20 always exist. A mode change re-selects which sub-block is *active*; the inactive block is held exactly, $\dot{x}_{inactive} = 0$, under a frozen inactive-state rule (PROJECT_DERIVED: continuity plus algebraic-residual minimisation). The integrator and the Jacobian therefore never resize. This is also why a later SG reclose is meaningful: nothing was structurally deleted at the trip.

The reset map R in (37) transfers *physical terminal conditions*, not state coordinates, and is owned by the device:

$$V_{bus} = y_{2b-1} + jy_{2b}, \quad I^- = I_{inj}(x^-, y, u, q^-), \quad (39)$$

$$S^- = V_{bus} \overline{I^-}, \quad P^- = \Re S^-, \quad Q^- = \Im S^-, \quad (40)$$

$$x^+|_{target} = \mathcal{I}_{q^+}(V_{bus}, P^-, Q^-), \quad x^+|_{inactive} = x^-|_{inactive}, \quad (41)$$

followed by the fail-closed acceptance test

$$|I_{inj}(x^+, y, u, q^+) - I^-| \leq 10^{-10}. \quad (42)$$

$\mathcal{I}_{q^+}$ is the target branch's own equilibrium/transfer initialiser. The invariant carried across the switch is therefore the terminal pair (V, I) — equivalently (P, Q) — and not the state vector. No PLL angle is copied out of the GFL branch, because the GFL branch owns no PLL state; the GFM angle is re-derived from V_{bus} and I^- . The bus index b is resolved from the device's registered bus position, so the map contains no hard-coded bus identity.

3.5 Why the recursion becomes an assignment, and what equation is solved

This subsection answers the most direct question: during a normal step we advance the recursion $x_{k+1} = x_k + \frac{h}{2}[f_q(x_k) + f_q(x_{k+1})]$, yet at a switching instant the code appears to simply *assign* $x^+ = (\text{a value})$. What equation is being solved, and what makes the two sides of the event sync with each other?

3.5.1 Why the recursion cannot be continued

Write the step that straddles the switching time t_e explicitly, with $t_k < t_e < t_{k+1}$. The trapezoidal rule requires

$$x_{k+1} - x_k = \frac{h}{2} \left[\underbrace{f_{q^-}(x_k, y_k)}_{\text{pre-switch mode}} + \underbrace{f_{q^+}(x_{k+1}, y_{k+1})}_{\text{post-switch mode}} \right]. \quad (43)$$

Equation (43) is *mathematically invalid*, because the trapezoidal rule is derived from $\int_{t_k}^{t_{k+1}} f dt$ on the assumption that f is *continuous* on that interval; at t_e the vector field jumps ($f_{q^-} \neq f_{q^+}$), so the order of accuracy collapses and the result approximates the integral of no field at all. The correct treatment is to **split the step at t_e** and treat t_e as the boundary of two subproblems: the one on $[t_k, t_e]$ uses f_{q^-} and the one on $[t_e, t_{k+1}]$ uses f_{q^+} , and each subproblem *needs its own initial condition*. The initial condition of the second subproblem is $x(t_e^+)$, which is exactly what the reset map must produce. The recursion is therefore not “gone” — it *ends one round and starts another*, with an initial-condition assignment in between.

3.5.2 The equation actually solved at the switching instant

x^+ is not a value chosen out of thin air; it is the *solution of an inverse problem* with an explicit governing equation. Let $C_q(x, V)$ be the terminal current-injection function of mode q . The problem is

$$\boxed{\text{find } x^+ \text{ such that } C_{q^+}(x^+, V_{bus}) - \underbrace{C_{q^-}(x^-, V_{bus})}_{= I^- \text{ already known}} = 0} \quad (44)$$

subject to the constraint that the inactive block keeps its values, $x^+|_{inactive} = x^-|_{inactive}$. Equation (44) is the equation that is really “solved”. In physical terms: *the controller may change its operating law instantly, but the current leaving the terminal must not jump*, because that current is tied to the energy stored in the filter inductances, which cannot change as a step without an infinite voltage. The continuity check (42) on $|I^+ - I^-| \leq 10^{-10}$ is therefore not a generic numerical quality check; it is **a verification that the inverse problem (44) was actually solved**. If the value exceeds the threshold, no consistent x^+ exists and the transaction must not commit.

3.5.3 The closed-form inversion: GFL→GFM step by step

The GFL→GFM case at 20 s is the best example because it is solvable *in closed form* with no iteration at all. Every step below is a direct back-substitution into the GFM branch equations.

Step 1 — read the terminal condition from the left side. Compute the current the GFL branch is actually injecting, using the GFL injection (22), and convert it to power:

$$I^- = \frac{(I_{p,reg}\gamma(|V|) - jI_{q,reg})e^{j\angle V}}{\kappa}, \quad S^- = V_{bus}\overline{I^-}, \quad P^- = \Re S^-, \quad Q^- = \Im S^-. \quad (45)$$

The three quantities (V_{bus}, P^-, Q^-) are *all* that is carried across the event; no GFL state variable is forwarded.

Step 2 — find the required internal voltage. The GFM branch injects current through the impedance $Z_{sys} = \kappa(R_e + jX_L)$ by (21), so for it to deliver exactly I^- the voltage behind the impedance must be

$$E_{terminal} = V_{bus} + Z_{sys}I^- \implies E_{VSM} = |E_{terminal}|, \quad \delta_{VSM} = \angle E_{terminal}. \quad (46)$$

This is the direct *inversion* of (21): originally we have E and find I ; now we have I and find E . The equation solves immediately because it is linear in I .

Step 3 — decompose E_{VSM}, δ_{VSM} back into state variables. We now have what the *combination* must be, and still have to split it into individual states. Start with the angle: the PLL is locked to the terminal-voltage angle ($\delta_{PLL} = \angle V_{bus}$, positive d -axis), so from $\delta_{VSM} = \delta_{PLL} + \delta_{IT}$,

$$\delta_{IT} = \text{wrap}_\pi(\angle E_{terminal} - \angle V_{bus}), \quad x_{PLL} = 0, \quad \Delta\omega_m = 0, \quad x_w = 0. \quad (47)$$

For the magnitude, invert the voltage-PI equation (15) to solve for the integrator:

$$\begin{aligned} E_{VSM} &= V_{ref} - m_q Q_f + k_{pv}(V_{ref} - V_f) + k_{iv}x_E \\ \implies x_E &= \frac{E_{VSM} - V_{ref} + m_q Q_f - k_{pv}(V_{ref} - V_f)}{k_{iv}}, \end{aligned} \quad (48)$$

which requires $k_{iv} > 0$ (the code checks this and fails closed otherwise). The five filter states are set exactly to the measured quantities at that instant ($P_{inv,f} = \kappa P^-, Q_{inv,f} = \kappa Q^-, V_{inv,f} = |V_{bus}|$,

and $I_{d,f}, I_{q,f}$ from projecting the current onto the PLL frame), because a freshly restarted filter must not create a spurious transient from an initial value that disagrees with its input.

Step 4 — verify the solution lies in the source-allowed region. Before accepting x^+ the code checks three layers and fails closed on each: $|I^-|$ must not exceed I_{max}^F/κ (if it did, the clamp would change P, Q , which is not acceptable), E_{VSM} must lie in $[E_{min}, E_{max}]$ of (17), and δ_{IT} must lie in $[\delta_{IT,min}, \delta_{IT,max}]$. It also rejects the case where $|I^-|$ lands exactly on the clamp boundary, because the derivative is discontinuous there and the solution is not unique.

Step 5 — verify by recomputing. Feed x^+ back into the GFM current equation and compare with I^- by (42). This confirms steps 2–3 contain no error. Theoretically the value must be exactly zero; the 10^{-10} tolerance therefore covers only floating-point round-off, not a physical relaxation.

3.5.4 Device-level sync versus system-level sync

The word “sync” is used at two levels with two different mechanisms.

Device level is steps 1–5 above: make the device itself consistent with the terminal condition at the *same* voltage V_{bus} . Note that throughout steps 1–5 the voltage V_{bus} is held fixed and never moves, so this is a per-device local problem that can be solved one device at a time.

System level is the right-limit solve (38), needed because when several devices change their laws at once (and/or the topology changes $Y_{q-} \rightarrow Y_{q+}$), the old voltage $y(t_e^-)$ is in general *no longer consistent* with the new KCL:

$$g_{q^+}(x^+, y(t_e^-), u) = Y_{q^+}V(t_e^-) - I_{bus}(x^+, y(t_e^-)) \neq 0. \quad (49)$$

One must therefore solve for $y(t_e^+)$ anew, *holding x^+ fixed*, by damped Newton

$$J_{yy} \Delta y^{(i)} = -g_{q^+}(x^+, y^{(i)}, u), \quad y^{(i+1)} = y^{(i)} + \lambda_i \Delta y^{(i)}, \quad J_{yy} = \frac{\partial g_{q^+}}{\partial y}, \quad (50)$$

where $\lambda_i \in (0, 1]$ comes from a backtracking line search that shrinks until the residual norm actually decreases (preventing over-long steps from diverging). It starts from $y^{(0)} = y(t_e^-)$ and stops when $\|g_{q^+}\|_\infty \leq$ the step tolerance; if it does not converge, the whole transaction rolls back.

The order **device first, then network** is essential and cannot be reversed: the device step needs a known V_{bus} to invert its equations, and the network step needs a known x^+ to assemble I_{bus} . The two steps are therefore one transaction that commits or rolls back together.

After this the system has a pair $(x(t_e^+), y(t_e^+))$ consistent with both the device equations and KCL, and the trapezoidal recursion restarts correctly with f_{q^+} as the vector field and this pair as the initial condition,

$$x_{k+2} - x(t_e^+) = \frac{h'}{2} [f_{q^+}(x(t_e^+), y(t_e^+)) + f_{q^+}(x_{k+2}, y_{k+2})], \quad h' = t_{k+2} - t_e. \quad (51)$$

In short, *the recursion never crosses the event; it is cut into two segments joined by a physical continuity condition, not by continuity of the state coordinates.*

3.5.5 The SG case: why an extra synchronism gate is needed

The IBR case above has a “sync” that solves immediately in closed form, because the converter may freely choose its internal angle (δ_{IT} is just a number in the state). The SG *cannot* do this: the rotor angle δ is tied to a real rotating mass and cannot change instantaneously. When closing

the SG breaker we therefore have *no freedom* to invert an equation for a desired δ^+ ; whatever δ happens to be must simply be accepted.

The roles are thus completely reversed: instead of *solving* for the continuity condition, we must *wait* until the physics brings the system close enough to continuity on its own. That is the job of the gate (54), which compares the two phasors on either side of the breaker and closes only when the difference is small enough in magnitude, frequency, and angle. The values $\Delta V = 0.0163$ pu, $\Delta f = 2.75 \times 10^{-6}$ pu, and $\Delta \theta = 3.76^\circ$ at 146.825 s are evidence that the condition was made true before closing, not enforced after.

This is the mathematical explanation of why the 145.000 s close request does not close immediately: *the breaker closure is an event with no supporting reset map*, so it must wait for the state to approach continuity by itself, whereas an IBR mode switch has a solvable reset map and happens the instant the transaction commands it.

3.6 The SG trip and the gated reclose

At the trip the SG states are *not* removed from x . The breaker-open branch replaces the connected-stator algebra with the open-circuit physics: $I_d = I_q = 0$, $T_e = 0$, the mechanical torque frozen at its pre-trip value, and open-circuit flux decay,

$$2H\Delta\dot{\omega} = T_m^{froz} - D\Delta\omega, \quad T_{d0}''\dot{E}_q'' = E_q' - E_q'', \quad T_{q0}''\dot{E}_d'' = E_d' - E_d'', \quad (52)$$

i.e. the $(X_d' - X_d'')I_d$ and $(X_q' - X_q'')I_q$ terms vanish with the stator current while $\dot{\delta} = \omega_0\Delta\omega$ continues. The rotor coasts, which is what makes a later resynchronisation physically meaningful.

The reclose is *gated*, not scheduled. A guard compares the network bus phasor against the SG open-circuit EMF $(E_q'' - jE_d'')e^{j\delta}$ and requires

$$|\Delta V| \leq 0.05 \text{ pu}, \quad |\Delta f| \leq 0.001 \text{ pu}, \quad |\Delta \theta| \leq 10^\circ, \quad (53)$$

expressed as the signed margin

$$m = \min(0.05 - |\Delta V|, 0.001 - |\Delta f|, 10^\circ - |\Delta \theta_{wrap}|) \geq 0, \quad (54)$$

sustained for a 0.5 s dwell. The requested close time is therefore only the earliest instant at which closure *may* occur: the request at 145.000 s produced an accepted close at 146.825 s.

4 Power-flow operating point

The power-flow is solved by the project Newton–Raphson routine. Q-limit enforcement is switched on, and the generator reactive-capability limits behind it are *source-defined*, not assumed or tuned: they are the IEEE14 generator data $Q_{\max}$ and $Q_{\min}$ in MVar (cases.case_matpower6_case14, mpc.gen columns 5 and 4), converted to per unit with the single system base $S_{\text{base}} = 100$ MVA,

$$Q_{\max, \text{pu}} = \frac{Q_{\max, \text{MVar}}}{100}, \quad Q_{\min, \text{pu}} = \frac{Q_{\min, \text{MVar}}}{100}, \quad (55)$$

with $P_{\max}$ taken the same way from mpc.gen column 9. Buses carrying no finite generator limit are left unconstrained ($Q_{\max} = +\infty$, $Q_{\min} = -\infty$) and print a dash.

It must be stated plainly that *enabling the mechanism is not the same as constraining this operating point*. The PV→PQ Q-limit switch is defined only for PV buses, and after the inverter

buses are mapped to PQ resources this case has no PV bus at all, so the switch resolves in zero rounds and changes nothing. Two consequences follow and are reported rather than masked: the slack bus 1 settles at $Q_g = -0.196$ pu against its registered $Q_{\min} = 0$, and bus 6 at $Q_g = 0.2476$ pu against $Q_{\max} = 0.24$. Neither is a solver error — a REF bus absorbs whatever reactive power closes the network balance and a PQ resource injects its scheduled Q — but neither is a limit the solve enforced. The limit columns therefore report registered source data, not satisfied constraints. The enforcement path itself is exercised by the baseline IEEE14 case, which does carry PV buses.

The admissible voltage band is $\pm 5\%$ about the 1.00 pu slack reference, i.e. $V_{\max} = 1.05$ pu and $V_{\min} = 0.95$ pu on every bus (CASE_DEFINED in case_ieee14bus_eecon49_switch). This is a reporting and admissibility band, not a PF equation input: the slack is the only voltage-controlled bus in this operating point, so the band does not act as a PV-setpoint clamp. One observation is reported rather than masked: bus 6 solves at 1.0575 pu, 7.5×10^{-3} pu above $V_{\max}$, because its mapped IBR6 reactive dispatch and the fixed 5–6 tap ratio are source/case-defined and were not adjusted to satisfy the band.

The bus roles (REF/PQ/GFL) in the Type column are the bus_role labelling descriptor (Section 1); being a label only, they do not alter (55) or any solved quantity.

Table 2: Bus power-flow results (pu)

Bus	Type	$ V $	θ (deg)	P_g	Q_g	P_L	Q_L	$Q_{\min}$	$Q_{\max}$	$P_{\max}$
1	REF	1.0000	0.0000	1.3319	-0.1957	0.0000	0.0000	0.00	0.10	3.32
2	GFL	0.9914	-3.2585	0.3172	0.1767	0.2170	0.1270	-0.40	0.50	1.40
3	GFL	0.9667	-8.6921	0.2884	0.1607	0.9420	0.1900	0.00	0.40	1.00
4	PQ	0.9832	-6.7064	0.0000	0.0000	0.4780	-0.0390	—	—	—
5	PQ	0.9858	-5.5221	0.0000	0.0000	0.0760	0.0160	—	—	—
6	GFL	1.0575	-6.7136	0.4443	0.2476	0.1120	0.0750	-0.06	0.24	1.00
7	PQ	1.0253	-6.9205	0.0000	0.0000	0.0000	0.0000	—	—	—
8	GFL	1.0494	-4.4090	0.2677	0.1491	0.0000	0.0000	-0.06	0.24	1.00
9	PQ	1.0219	-8.6416	0.0000	0.0000	0.2950	0.1660	—	—	—
10	PQ	1.0202	-8.5886	0.0000	0.0000	0.0900	0.0580	—	—	—
11	PQ	1.0347	-7.7782	0.0000	0.0000	0.0350	0.0180	—	—	—
12	PQ	1.0411	-7.6862	0.0000	0.0000	0.0610	0.0160	—	—	—
13	PQ	1.0342	-7.8345	0.0000	0.0000	0.1350	0.0580	—	—	—
14	PQ	1.0087	-9.3374	0.0000	0.0000	0.1490	0.0500	—	—	—

Table 3: From-end line flows and losses (pu)

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	0.910918	-0.153112	0.016392	-0.002303
2	1	5	0.420987	-0.042553	0.009593	-0.008907
3	2	3	0.466697	0.013221	0.010470	0.002115
4	2	4	0.316733	-0.064476	0.006065	-0.014738
5	2	5	0.211276	-0.049810	0.002649	-0.025732
6	3	4	-0.197333	-0.018164	0.002803	-0.005014
7	4	5	-0.448842	0.084724	0.002882	0.009089
8	4	7	0.018419	-0.096315	0.000000	0.001990
9	4	9	0.062955	-0.012297	0.000000	0.002223
10	5	6	0.092297	0.001911	0.000000	0.001920
11	6	11	0.130821	0.059907	0.001758	0.003682
12	6	12	0.086088	0.027130	0.000895	0.001864
13	6	13	0.207667	0.085524	0.002983	0.005875
14	7	8	-0.267650	-0.134127	0.000000	0.015018
15	7	9	0.286069	0.035823	0.000000	0.008698
16	9	10	-0.003258	0.021664	0.000015	0.000039
17	9	14	0.057282	0.023368	0.000466	0.000991
18	10	11	-0.093273	-0.036375	0.000790	0.001849
19	12	13	0.024192	0.009266	0.000137	0.000124
20	13	14	0.093739	0.030791	0.001556	0.003168

5 200-s event and mode evidence

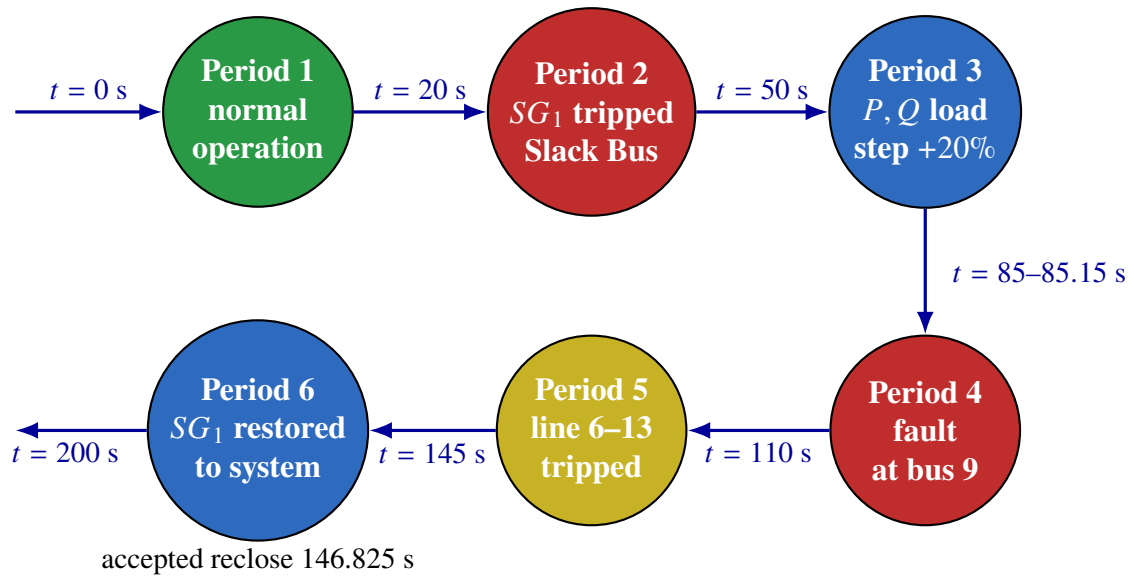


Figure 2: Six-period disturbance sequence of the 200-s simulation from the accepted production event log: normal operation at 0 s, SG trip at 20 s, P, Q load step +20% at 50 s, bus-9 fault 85–85.15 s, line 6–13 trip at 110 s, and SG restoration at 145 s (accepted reclose 146.825 s).

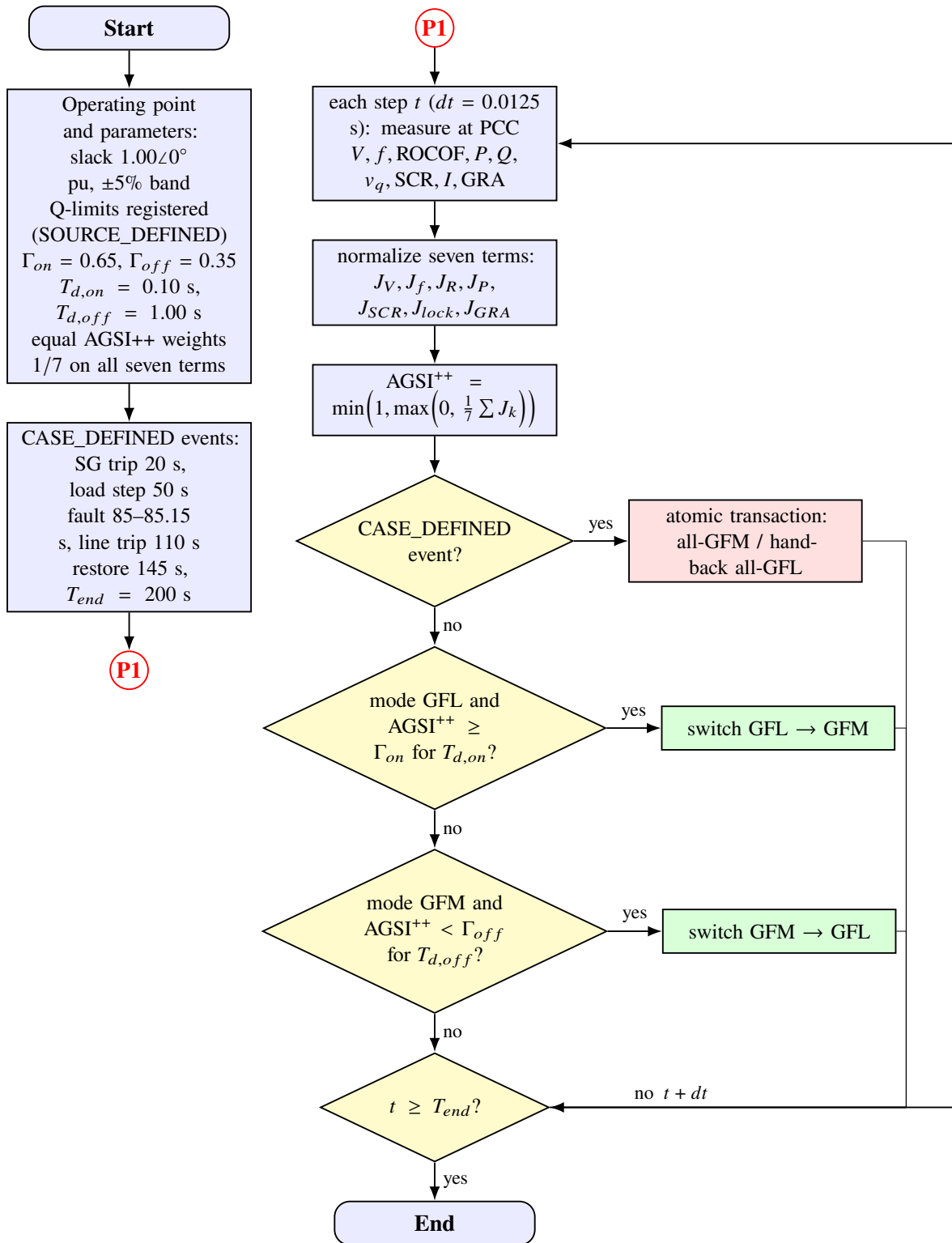


Figure 3: GFL/GFM switching-strategy flowchart. The left column sets the operating point, parameters, and CASE_DEFINED events; the right column is the online loop that measures the PCC signals each step, forms the seven-term AGSI++ of (3) actually implemented by `ibr.SwitchableIbr6`, and switches mode by threshold and dwell or through the atomic CASE_DEFINED transaction.

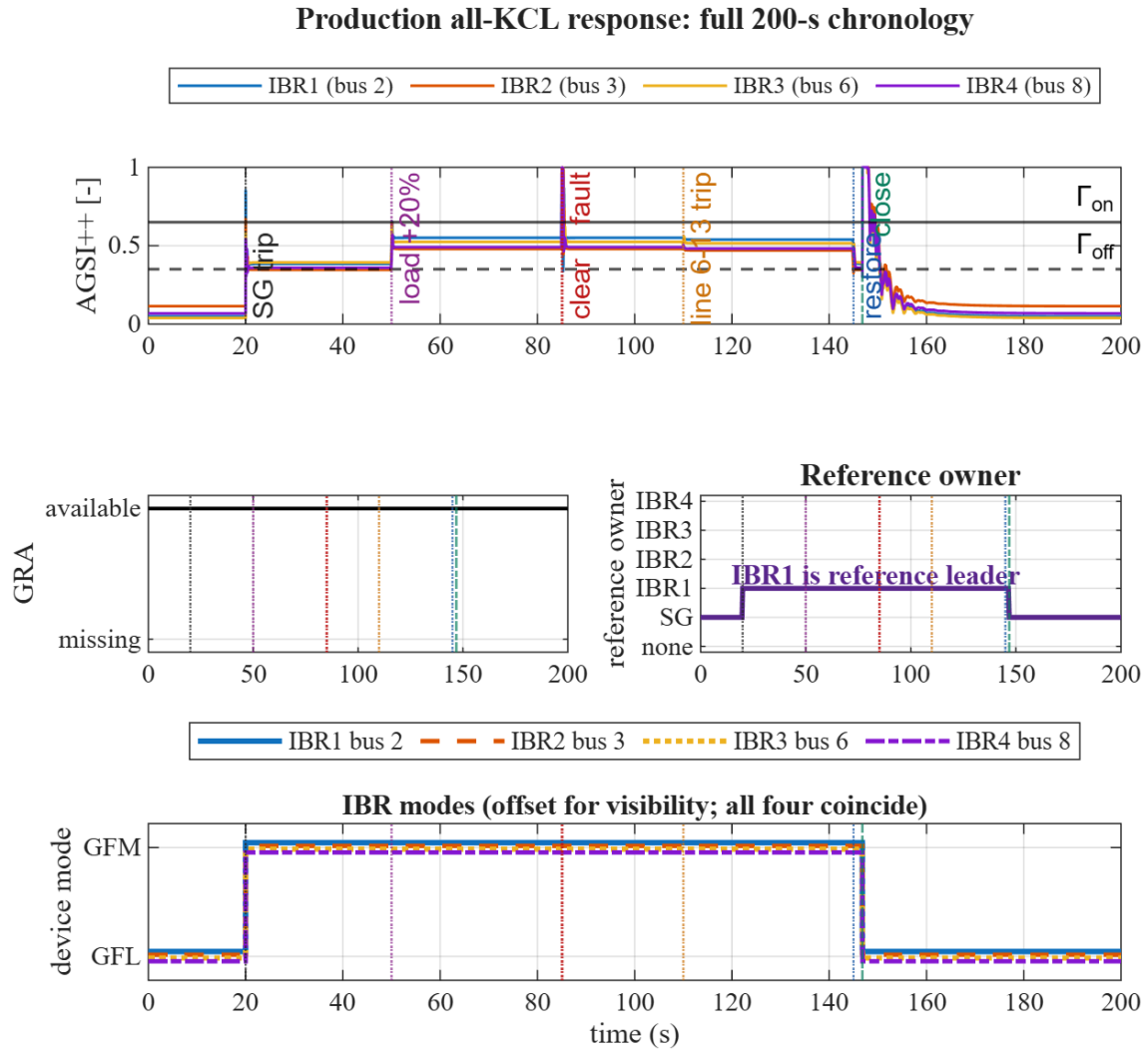


Figure 4: Bounded AGSI++, GRA, reference ownership, and the combined IBR1–IBR4 mode timeline. The owner trace identifies IBR1 at bus 2 as the grid-forming angle/frequency leader while the SG is open. All IBRs are GFL before 20 s, GFM until the accepted handback, and GFL thereafter; the four mode traces are drawn with distinct line styles and a small display-only vertical offset (± 0.045 of the 0/1 mode coordinate) because they coincide exactly.

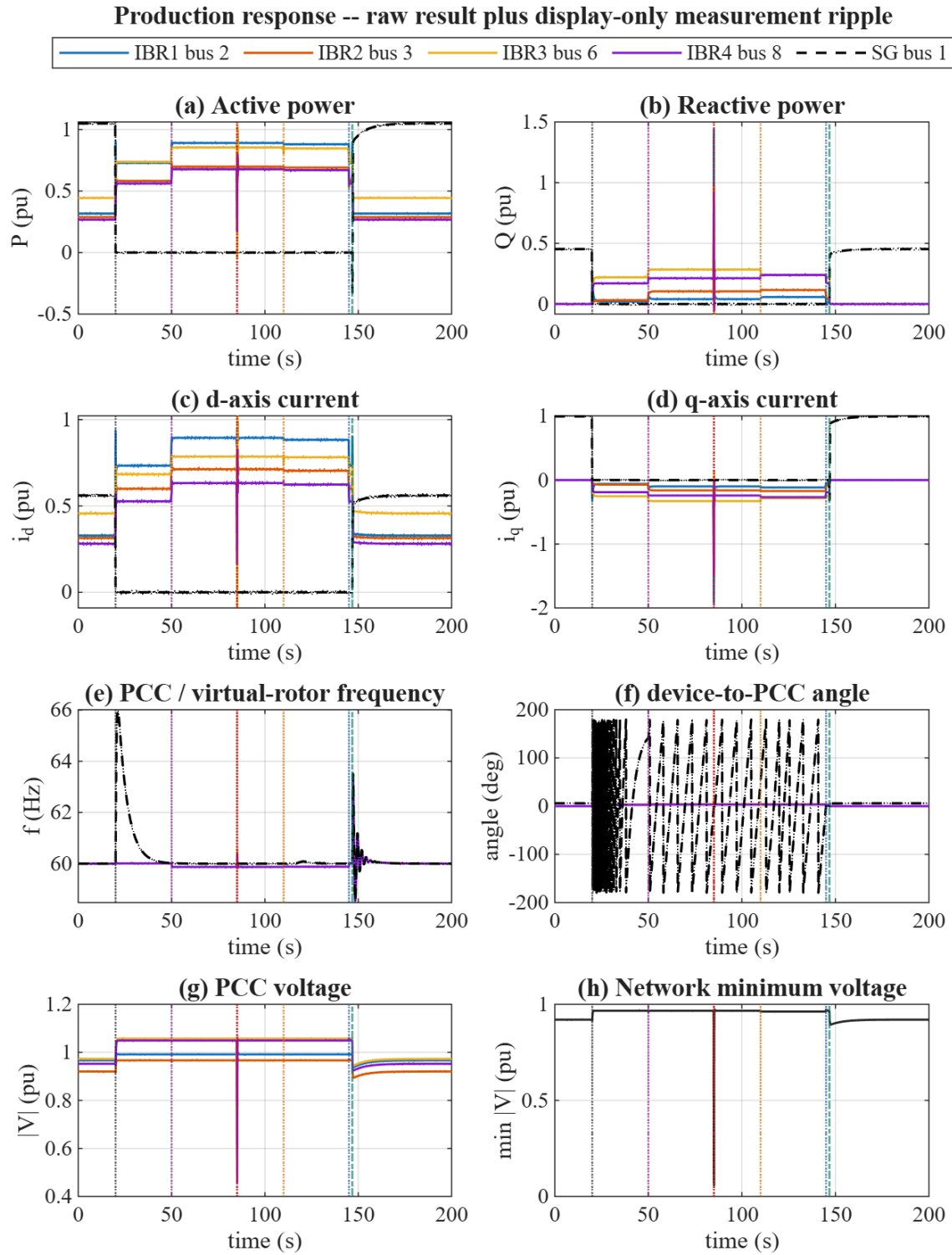


Figure 5: IBR and SG active/reactive power, frame currents, frequency, device-to-PCC angle, PCC voltage, and network minimum voltage. Solid traces are raw production results; dotted band-limited ripple is display-only and does not enter any equation or decision. Remaining vertical event excursions belong to the raw switching transient, not to the ripple.

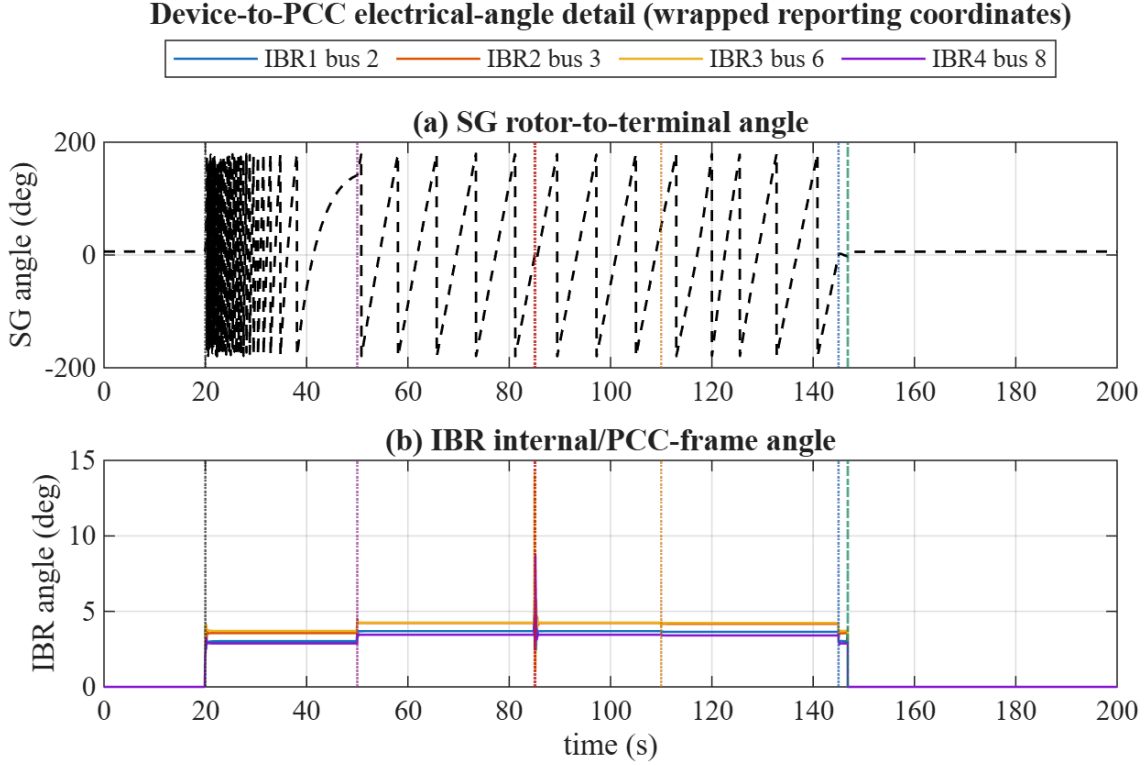


Figure 6: Angle detail. The offline SG rotor slips relative to the energized grid; the plotted coordinate is wrapped to $[-180, 180)$ degrees.

At the accepted close, the synchronism mismatch is $\Delta V = 0.0163$ pu, $\Delta f = 2.75 \times 10^{-6}$ pu, and $\Delta \theta = 3.76^\circ$; all three lie inside the guard limits (0.05 pu, 0.001 pu, 10°). At the 200 s endpoint the all-bus voltage range is 0.9203–1.0000 pu: the SG and IBR regulating buses sit inside the $\pm 5\%$ band, but five remote load buses solve below $V_{\min} = 0.95$ pu. SG electrical power is 1.0497 pu, reactive output is 0.4531 pu, and SG frequency is 60.0002 Hz, i.e. settled from the transient excursion. The trajectory therefore demonstrates a successful declared handback and frequency recovery, while the endpoint reactive profile shows the new operating point has not yet returned every bus to the $\pm 5\%$ band.

6 Evidence boundary and reproduction

The primary run uses $\Delta t = 0.0125$ s. The accepted-close time (146.825 s) is an event-landed output of the synchronism guard, not a scheduled constant, so event-time step-independence is not claimed. The run proves numerical execution of the declared chronology, atomic mode transfer, full KCL, synchronism-gated close, and return to all GFL. It does not validate the diagnostic synchronizer, governor parameters, energy reserve, or a general protection design against field measurements. The dotted display ripple is never used as model evidence; the solid raw traces remain visible.

```
pf_init_paths;
addpath(fullfile('scripts','reporting'));
generate_ieee14_switch_report_figures(reuse_cache=true);
```

All production solvers are project-owned MATLAB. The targeted numerical, producer and consumer, hybrid event, fail-closed, and external-solver-prohibition suites pass 108/108 tests. The

full repository regression was intentionally omitted under the repository risk policy because these targeted gates cover the changed producer, consumers, and failure paths.